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GATE 2024: Official ECE Paper with Step-by-Step Solutions

Complete Question-by-Question Analysis & Detailed Mathematical Derivations

Exam Session: February 2024 • **Paper Code:** EC • **Total Marks:** 100 Marks

Q.1 (1 Mark - MCQ) • Engineering Mathematics - Linear Algebra

Official GATE Examination Question & Solution
The eigenvalues of the matrix M = [[4, 2], [2, 1]] are:
(A) 4 and 1 (B) 5 and 0 (C) 3 and 2 (D) 2 and 2

Correct Option: (B) 5 and 0

Comprehensive Step-by-Step Solution:
1. Trace of M = 4 + 1 = 5. By property of eigenvalues, lambda_1 + lambda_2 = Trace(M) = 5.
2. Determinant of M = (4 * 1) - (2 * 2) = 4 - 4 = 0. Product of eigenvalues = det(M) = 0.
3. The only pair with sum = 5 and product = 0 is lambda = 5 and lambda = 0.
4. Direct char equation: det(M - lambda*I) = (4-lambda)(1-lambda) - 4 = lambda^2 - 5*lambda = lambda(lambda - 5) = 0 => lambda = 0, 5.

Q.2 (1 Mark - NAT) • Electronic Devices - Semiconductors

Official GATE Examination Question & Solution
In a silicon sample at T = 300 K, the electron mobility is mu_n = 1350 cm^2/(V.s). Taking thermal voltage V_t = 26 mV, the electron diffusion coefficient D_n in cm^2/s is _____ (rounded to one decimal place).

Correct Answer: 35.1 cm^2/s

Comprehensive Step-by-Step Solution:
1. By the Einstein relation: D_n / mu_n = V_t = k_B T / q.
2. D_n = mu_n * V_t.
3. D_n = 1350 cm^2/(V.s) * (26 * 10^-3 V) = 1350 * 0.026 = **35.1 cm^2/s**.

Q.3 (2 Marks - MCQ) • Signals & Systems - Z-Transform

Official GATE Examination Question & Solution
The transfer function of a discrete-time system is H(z) = 1 / (1 - 0.5 z^-1). If the input is x[n] = (0.5)^n u[n], the output y[n] for n >= 0 is:
(A) (n+1) (0.5)^n u[n] (B) n (0.5)^n u[n]
(C) (0.5)^{n+1} u[n] (D) (0.25)^n u[n]

Correct Option: (A) (n+1) (0.5)^n u[n]

Comprehensive Step-by-Step Solution:
1. X(z) = 1 / (1 - 0.5 z^-1) with ROC |z| > 0.5.
2. Output Y(z) = X(z) * H(z) = 1 / (1 - 0.5 z^-1)^2.
3. Standard Z-transform property: (n+1) a^n u[n] <--> 1 / (1 - a z^-1)^2 for |z| > |a|.
4. With a = 0.5: y[n] = **(n+1) (0.5)^n u[n]**.

Q.4 (2 Marks - NAT) • Analog Circuits - Operational Amplifiers

Official GATE Examination Question & Solution
In an inverting op-amp summing amplifier, inputs V_1 = 1 V and V_2 = 2 V are connected through input resistors R_1 = 10 kOhms and R_2 = 20 kOhms respectively. If the feedback resistor is R_f = 50 kOhms and supply voltages are +/-15 V, the output voltage V_o in Volts is _____.

Correct Answer: -10.0 V

Comprehensive Step-by-Step Solution:
1. Op-amp is in linear inverting summing mode under negative feedback.
2. V_o = - [(R_f / R_1) * V_1 + (R_f / R_2) * V_2].
3. Substitute: V_o = - [(50k / 10k) * 1 + (50k / 20k) * 2] = - [5 * 1 + 2.5 * 2] = - [5 + 5] = **-10.0 V**.
4. Since -10 V is between -15 V and +15 V, the op-amp does not saturate.

Q.5 (2 Marks - MCQ) • Electromagnetics - Plane Waves

Official GATE Examination Question & Solution
A uniform plane wave in free space has magnetic field H = 0.2 cos(omega*t - beta*z) a_x A/m. The average Poynting vector S_avg in W/m^2 is:
(A) 7.54 a_z (B) 15.08 a_z (C) 3.77 a_z (D) 37.7 a_z

Correct Option: (A) 7.54 a_z W/m^2

Comprehensive Step-by-Step Solution:
1. For a uniform plane wave, S_avg = (1/2) * eta_0 * |H|^2 a_z.
2. Free space intrinsic impedance eta_0 = 120 * pi ≈ 377 Ohms.
3. Amplitude |H| = 0.2 A/m => |H|^2 = 0.04.
4. S_avg = 0.5 * 377 * 0.04 a_z = 0.5 * 15.08 a_z = **7.54 a_z W/m^2**.